Advanced Microeconomics

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Problem Set 8: Mathematical solutions

1 Application

1) A owns a car worth 50,000 EUR which can get stolen with probability 1%. She can purchase insurance coverage of the amount $q \in [0; 50,000]$ at premium $\pi = 0.05$ EUR for each Euro covered. Her utility function is $u = ln(w)$. Assume she has no other assets.

a) Set up her maximization problem to decide on the amount of insurance, q, she wants to buy.

> **Solution:**
> $$\text{max: } EU = 0.99 \ln(50,000 - 0.05q) + 0.01 \ln(q - 0.05q)$$

b) How much insurance will she choose to buy?

> **Solution:**
> $q = 10.000$

c) How much profits does the insurance company earn on insuring A?

> **Solution:** Profit = 400 EUR.

e) How much insurance will she buy if insurance companies charge an actually fair insurance premium?

> **Solution:** $q^* = 50,000$.

2) Consider an insurance company that does not have the inclination to tailor contracts specifically to individuals. Instead, it will offer a "standard contract" with the premium $r = 100$a€nd payout $q = 500$i€n case the person gets sick to anyone who will purchase it.

a) A has healthy-state income I_H of 500a€nd sick-state income I_S, of 0.€ He has a probability of illness $p = 0.1$. Is the standard contract fair and/or full for A? If he ends up getting sick, what will his final income be?

Solution: To Individual A the contract is unfair and full. The expected income is 400€.

b) B has $I_H = 500$ € and $I_S = 0$, but his probability of illness $p = 0.2$, is higher than A's. Is the standard contract fair and/or full for B? How does purchasing the standard contract affect B's expected income?

Solution: To individual B the contract is fair and full. The expected income is 400€.

c) C has $I_H = 1,000$ € and $I_S = 0$ €, with probability of illness p = 0.2. Is the standard contract fair and/or full for C?

Solution: To individual C the contract is fair and partial.

d) Suppose there is a customer D for whom the standard contract is partial and actuarially unfair in the insurance company's favor. Give a set of possible values for D's I_H, I_S, and p. Recall that we always assume $I_H > I_S$.

Solution: The contract is unfair as long as $p \neq 0.2$. The contract will represent less than full insurance whenever $q < I_H - I_S$.

e) Now suppose that we have learned that D's $I_S = 200$ €, but we do not know the value of his healthy-state income, or his probability of falling ill. Derive an upper bound for p and a lower bound for I_H.

Solution: $I_H > 700$ € and $p < 0.2$.

3) Consider an individual whose utility function over income I is $U(I)$, where U is increasing smoothly in I and is concave. Let $I_S = 0$ be this person's income if he is sick, let $I_H > 0$ be his income if he is healthy, let p be his probability of being sick, let $E[I]$ be expected income, and let $E[U]$ be his expected utility when he has no insurance.

a) Write down algebraic expressions for both E[I] and E[U] in terms of the other parameters of the model.

Solution:

$$E[I] = (1-p)I_H + pI_S = (1-p)I_H$$

$$E[U] = (1-p)U(I_H) + pU(I_S)$$

c) Derive an algebraic expression for the consumer surplus M. Hint: You may assume that U is invertible and its inverse is U^{-1}.

Solution:

$$M = (1-p)I_H - U^{-1}((1-p)U(I_H)).$$